A1 Code structure

Functions, type hints, docstrings, naming, config files and PEP 8.

A1-L1 Single-purpose functions and clear names

Each strong function passes the one-sentence test. The weak one needs "and" four times.

The five jobs a pipeline function does, one at a time	Read, transform, validate, write, log. An orchestrator calls them in order.
The one-sentence test	Describe the function without using "and". If you need "and", split it.
Function, class and constant naming	check_table_health, CheckResult, MAX_WEIGHT_CV.
What does a get_ or check_ prefix promise?	No side effects. The function reads and returns. It never writes or deletes.

A1-L2 Type hints and docstrings a reviewer can trust

The strong version tells mypy, the reviewer and the caller the same story. The weak version tells three different ones.

```python
def weighted_share(df: DataFrame, flag_col: str) -> float | None:
    """Weighted share of rows where flag_col is 1.

    Args:
        df: Respondents with final_weight.
        flag_col: 0/1 column to share.

    Returns:
        Share between 0 and 1, or None when the total weight is 0.

    Raises:
        ValueError: If flag_col is missing from df.
    """
```

Optional parameter that defaults to None	param: str | None = None. Never param: str = None.
A fixed set of string values	Literal["PASS", "WARN", "FAIL"] from typing.
Google docstring sections	One-line summary, then Args, Returns, Raises.
Why merge_keys: list[str] and not list?	mypy then rejects a string, which Python would split into characters.

A1-L3 Config over hardcoded values, and PEP 8 with ruff

The strong version runs unchanged in dev and prod. The contract file decides the table and the limits.

What belongs in configuration?	Anything that changes between environments, trackers or contract versions: names, maps, thresholds, rules.
When do you validate configuration?	Once, on load, before any data moves. Fail fast with the key's name.

Why not raw.get("catalog", "research_prod")?	A default hides a missing key and can point a run at production.
The two ruff commands for CI	ruff check . and ruff format --check .